Line (GAC): I ASPHALT PAVER OPERATION

Competency: I5 Perform paving machine maintenance

Objectives

To be competent in this area, the individual must be able to perform regular daily maintenance on the paving machine.

LEARNING TASKS	CONTENT
1. Service lubrication system (greasing)	• Identify and lube all points • Load grease gun
2. Service air intake system as required (air filters/pre cleaners)	• Restriction indicators • Pre cleaner service • Cleaning vs. replacement, check employer policy (some insist on never using air – replace only) • Use air only with reduced pressure
3. Drain fuel tank sump, water separator fuel filters if equipped	• Drain, assess • Water, debris, scale, etc.
4. Add fluids as required	• Identify correct fluids for each component • Add fluids as required • Avoid contamination with dirt, water • Describe two-sided dipsticks with levels for stopped/idling (Cat engines, transmissions)
5. Inspect and clean components as required	• Radiator and oil cooler • Filter plugs, dipsticks • Battery terminals
6. Perform housekeeping tasks	• Garbage • Excess grease • General clean of screed, hoppers, augers, screed decks, extensions, top of pavers, etc. • Methods • Scraping • Release agents • Environmental concerns • Biodegradable solvents • Commonly missed spots

Achievement Criteria

Performance	The learner will perform regular daily maintenance on the asphalt paver. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will demonstrate all elements of daily maintenance.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist.

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I6 Comply with markers, grades and stakes

Objectives

To be competent in this area, the individual must be able to describe and apply job control and basic engineering knowledge during practical training.

LEARNING TASKS	CONTENT
1. Describe abbreviations used on stakes	• B/M - bench mark • C - cut • C/B - catch basin • SMH - sanitary manhole • Elv or EL - elevation • F/G - finish grade • P/L - property line • R/W - right of way • San or S - sanitary • Mon - monument • G - gas • PVC - poly vinyl chloride
2. Interpret stakes	• Property line, iron pins • Centre line, stationinng • Offset • Cut, fill • Grade • Slope • Reference points • Bench mark, temporary bench marks: geodetic and referenced to assumed elevations • Survey monuments: geodetic, precise, permanent, protected by law • Cross-heads • Location of works, e.g., manholes, catch basins, headwalls, etc.
3. Describe fundamentals of paver operation	• Follow the line • Precision developed through expereince • Aim vs. steering – set sight ahead • Pay attention! • Trucks come in straight

- Keep it smooth

Achievement Criteria

Performance
Learner will demonstrate job control and basic engineering knowledge.
Learner will demonstrate safe practices during practical training and on the job.

Conditions
The learner will apply information from the stakes to determine:

- C - cut
- C/B - catch basin
- SMH - sanitary manhole
- Elv or EL - elevator
- F/G - finish grade
- G - gas
- Centre line, stationing
- Offset
- Cut, fill
- Grade
- Slope

Criteria
The learner will be capable of maintaining:

- C - cut
- C/B - catch basin
- SMH - sanitary manhole
- Elv or EL - elevator
- F/G - finish grade
- G - gas
- Centre line, stationing
- Offset
- Cut, fill
- Grade
- Slope

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I7 Operate asphalt paver

Objectives

To be competent in this area, the individual must be able to perform standard operating functions.

LEARNING TASKS	CONTENT
1. Perform standard moves	• Move forward, stop, back up, stop • Apply park brake, lockout transmission
2. Maneuvering the paver	• Hills and slopes • Forwards and backwards
3. Position paver	• Efficient positioning of paver pre-operational • Move into appropriate position for several scenarios
4. Simulate paving laying down a mat using 5mm crush and sand mixture or similar compact quality material or asphalt	• Move over varied conditions, choose speeds appropriate, choose pattern

Achievement Criteria

Performance	The learner will demonstrate standard operating functions of the asphalt paver. Learner will demonstrate safe operating functions during practical training and on the job.
Conditions	The learner will work with the haul truck operator, ground crew and screed operator to safely operate the paver. Learners within the in-school technical training will be expected to complete tasks on flat surfaces only. Learners on the job will be required to operate the asphalt paver on various grades, slopes and in various conditions.
Criteria	To complete the in-school practical lab requirement for the paver, the learner must be able to demonstrate the following practical skills:

- Demonstrate the safe operation of equipment including observation of surroundings
- Perform pre-start checks, start-up/shut-down procedures and monitor performance of the equipment
- Perform daily maintenance tasks
- Perform standard operating procedures with paver including:
    - Move forward, stop, back up, stop (flat elevation)
    - Apply park brake, lock out transmission
    - Shift transmission
    - Open and close hopper
    - Set controls for rate of asphalt flow
    - Position paver into appropriate position to start paving
    - Guide haul truck into position to offload
    - Lay asphalt determined by site, equipment and conditions
    - Follow the line
    - Demonstrate proper hand signals to haul truck operator
    - Communicate with screed operator and ground crew
    - Demonstrate proper cleaning techniques for hopper, conveyor, etc.
- To complete the job-site practical requirement for the paver, the learner must be able to demonstrate all of the above practical skills on various grades, slopes and in various conditions.
- Observe learners performing learning tasks and assess competency using a rating scale.
- Time and quantity as instructor deems appropriate visual checks, constant awareness of space, position, clearance, safe movement.
- Operate with smooth and precise control of functions, appropriate gear and engine speed.
- Enter and exit machine safely (3 point contact, parking position, brake).

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I8 Follow shut down procedures

Objectives

To be competent in this area, the individual must be able to perform correct shut-down procedures for the paving machine.

LEARNING TASKS	CONTENT
1. Perform shut-down procedures	• Position that allows access in case of mechanical trouble at start up • Safe parking position, set parking brake • Level position for checking of fluids • Cool down before shut down • Walk around, look for leaks, general check • Secure machine, locks, etc.

Achievement Criteria

Performance	The learner will demonstrate correct shut down procedures for the asphalt paver.
Conditions	The learner will follow shutdown procedure as listed by the manufacturers guidelines.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I9 Transport paver

Objectives

To be competent in this area, the individual must be able to understand the correct transport procedures for the paving machine.

LEARNING TASKS	CONTENT
1. Describe compliance with wide load regulations	• Signage • Pilot car if applicable • Removing extension
2. Describe roles and responsibilities	• Truck driver is usually responsible for transporting equipment
3. Describe how to prepare machine for transport	• Remove attachments • Secure equipment • Ensure all guards are in place
4. Describe loading and unloading procedures	• Operator is not usually involved in loading/unloading • Use of tie-downs • Safe movement of maching on/off trailer

Achievement Criteria

Performance	The learner will demonstrate safe practices while preparing the asphalt paver for transport. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will prepare the machine for transport using appropriate tools and equipment to manufactures specification.
Criteria	The learner will be able to secure and prepare a machine for transport within a given time frame.

Line (GAC): J ASPHALT SCREED OPERATION

Competency: J1 Describe paving screed and attachments

Objectives

To be competent in this area, the individual must be able to describe the functions of the screed and the attachments used for the screed process.

LEARNING TASKS	CONTENT
1. Describe screed categories	• Rigid • Variable
2. Describe common attachments	• Extension, hydraulic wings, automatic grade controls (laser) • String line vs. boom vs. ski boom • Cut off shoe to narrow down joint, curb form
3. Describe principal functions	• Basic mix thickness control • How material gets to screed • Attachments • Different circumstances that require adjustments • Timing

Line (GAC): J ASPHALT SCREED OPERATION

Competency: J2 Apply screed work safety procedures

Objectives

To be competent in this area, the individual must be able to:

- Describe how to work safely while working on and around a screed. This will include understanding key regulations, machine limitations, operating procedures and tasks and the use of PPE's.
- Comply with safety requirements at all times during this program.

LEARNING TASKS	CONTENT
1. Interpret job plan	• Communicate job plan • Path • Pattern • Asphalt thickness • Temperature • Crown • Obstacles
2. Describe machine-specific safety hazards	• Dust, pressure (machine malfunctions/ kickbacks) blind spots, fumes, hydraulic lines, noise • Block outs • Heat related injuries • Proper work boots (steel or composite) • Burnt foot has happened (in a day and in cold weather when boots get wet) • Proper hydration
3. Describe traffic-control requirements	• Number of TCP's required • Signage • Signals • Communication
4. Describe proper PPE	• Workers are responsible to provide general purpose gloves, safety footwear, hard hat - employers are responsible for providing all other safety items • Employers must ensure that PPE is properly worn when required, and is cleaned, inspected, maintained • Employers must ensure that workers are adequately trained: correct use, limitations, maintenance

LEARNING TASKS	CONTENT
5. Describe lockout procedures	• Fuel supply • Battery lockout, etc.
6. Describe how to secure machine for safety	• Vandal proof take ancillary parts off • Park safely block elevated equipment, use parking brake and chocks

Achievement Criteria

Performance	The learner will demonstrate safe work practices while working with and operating the asphalt screed. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will: • Use correct PPE • Demonstrate machine specific safety awareness
Criteria	The learner will demonstrate correct lock-out procedures Secure the machine for safety including: • Safety locks • Blocks

Line (GAC): J ASPHALT SCREED OPERATION

Competency: J3 Describe asphalt screed operator roles and responsibilities

Objectives

To be competent in this area, the individual must be able to understand the roles and responsibilities of the asphalt screed crew.

LEARNING TASKS	CONTENT
1. Describe how to plan work procedures	• Requirements for EPS
2. Describe how to maintain mat thickness	• Width and volume control of asphalt mix
3. Describe communication techniques	• Paver operator and raker man
4. Describe problem solving strategies	• Anticipate adjustments and timing effect (several feet)
5. Describe the importance of teamwork	• Paver and screed operator
6. Describe importance of adhering to standards	• Maintain and correct volume – not under or over

Line (GAC): J ASPHALT SCREED OPERATION

Competency: J4 Perform screed maintenance

Objectives

To be competent in this area, the individual must be able to perform regular daily screed maintenance.

LEARNING TASKS	CONTENT
1. Service lubrication system (greasing)	• Identify and lube all points • Load grease gun • Lubricate thickness control “screws”
2. Inspect and clean components as required	• Radiator and oil cooler • Filter plugs, dipsticks • Battery terminals
3. Perform housekeeping tasks	• Garbage • Excess grease • Clean windows, sweep out cab, general cleanliness
4. Maintain screed plate	• Maintaining screed plate and adjust • Prolem: scuff mark on finished product • Problem: more wear • Could reveal themselves in a day

Achievement Criteria

Performance	The student will perform regular daily maintenance on the asphalt screed. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will demonstrate all elements of daily maintenance.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist.

Line (GAC): J ASPHALT SCREED OPERATION

Competency: J5 Operate asphalt screed

Objectives

To be competent in this area, the individual must be able to perform basic operating functions during practical training.

LEARNING TASKS	CONTENT
1. Set-up screed	• Thickness control – set up screed for thickness • Make proper adjustments
2. Maintain proper thickness	• Maintain and correct volume – not under or over
3. Check depth of mat	• Sing a poker: a stick or piece of metal

Achievement Criteria

Performance
The learner will safely demonstrate standard operating functions
Learner will demonstrate safe practices during practical training and on the job.

Conditions
To complete the practical requirement for the screed, the learner must be able to demonstrate the following practical skills:

- Demonstrate the safe operation of equipment including observation of surroundings.
- Perform pre-operational checks and monitor performance of the equipment.
- Perform daily maintenance tasks

Criteria
Perform standard tasks with equipment including:

- Set controls for proper mat thickness
- Lower screed onto blocks, allowing for compaction
- Operate screed heating system (propane/electric/diesel as equipped)
- Coordinate with paver operator to ensure appropriate speed
- Make adjustments to angle of attack, contour, etc.
- Monitor hydraulic systems
- Operate ancillary equipment if practical – e.g., lights

Line (GAC): J ASPHALT SCREED OPERATION

Competency: J6 Follow shut down procedures

Objectives

To be competent in this area, the individual must be able to perform correct shut down procedures for the screed.

LEARNING TASKS	CONTENT
1. Perform shut down procedures	• Position that allows access in case of mechanical trouble at start up • Level position for checking of fluids • Walk around, look for leaks, general check • Secure machine, locks, etc.
2. Clean screed after use	• Proper cleaning to prevent build up • Coordinate with paver to clean both machines simultaneously • Solvents and scraping (same for paver and in tandem) 20 min

Achievement Criteria

Performance	The learner will demonstrate correct shut down procedures for the asphalt screed.
Conditions	The learner will follow shutdown procedure as listed by the manufacturers guidelines.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist

Line (GAC): K RAKING OPERATION

Competency: K1 Describe raker roles and responsibilities

Objectives

To be competent in this area, the individual must be able to understand the roles and responsibilities of each of the team members of a asphalt paver crew.

LEARNING TASKS	CONTENT
1. Describe typical raker duties	• Does prep where machines can't go • Smaller jobs: manhole levelling asphalt mix to conform to grade appurtenances • Repair and level mistakes (made by screed or paver) • Finishing a pass, removing excess and squaring off edges • Raking joints • Longitudinal or tranverse centerline joints

Line (GAC): K RAKING OPERATION

Competency: K2 Apply raking safety procedures

Objectives

To be competent in this area, the individual must be able to:

- Describe how to work safely while working on and around paving equipment. This will include understanding key regulations, machine and personal limitations and the use of PPEs.
- Comply with safety requirements at all times during this program.

LEARNING TASKS	CONTENT
1. Interpret job planning	• Communicate job plan • Path • Pattern • Asphalt thickness • Temperature • Crown • Obstacles
2. Describe raker-specific safety hazards	• Dust, pressure (machine malfunction/kickback) blind spots, fumes, hydraulic lines, noises • Block outs • Heat related injuries • Proper work boots (steel or composite) • Burnt foot has happened (in a day and in cold weather when boots get wet) • Proper hydration is essential • Main focus is on awareness of other equipment, i.e., Compaction Rollers and vehicle traffic • Rake handles are long – keep handles out of traffic
3. Describe traffic control requirements	• Signage • Signals • Communication

LEARNING TASKS	CONTENT
4. Describe proper PPE	• Workers are responsible to provide general purpose gloves, safety footwear, hard hat - employers are responsible for providing all other safety items • Employers must ensure that PPE is properly worn when required, and is cleaned, inspected, maintained • Employers must ensure that workers are adequately trained: correct use, limitations, maintenance

Achievement Criteria

Performance	The learner will demonstrate safe work practices while working with and around the machines on site. Learners will demonstrate safe practices during practical training and on the job.
Conditions	The learner will: • Use correct PPE • Demonstrate machine specific safety awareness • Apply ergonomic safety awareness
Criteria	The learner will: • Demonstrate correct use of PPE's • Communicate with the traffic control people • Work safely around equipment • Apply ergonomic safety awareness

Line (GAC): K RAKING OPERATION

Competency: K3 Place asphalt hot mix

Objectives

To be competent in this area, the individual must be able to perform basic raking functions and techniques during practical training.

LEARNING TASKS	CONTENT
1. Perform common hand work crew duties	• Raker man on any size crew • Prep road • Intersection • Drive ways • Trench patching • Pothole filling • Grade deviations to minimize differential compaction • Squaring off end of the run/mat
2. Demonstrate common techniques	• Bumping • Ramping • Working around manholes

Achievement Criteria

Performance	The learner will apply raking techniques to produce a quality mat.
Conditions	The learner will use rake, lute, and tamping equipment to demonstrate the correct placement of hot mix asphalt. The learner will demonstrate various techniques with regard to the placement of hot mix asphalt.
Criteria	The learner will perform basic raking and hand work tasks including: • Apply coals or simulated emulsified asphalt to joints prior to laying asphalt • Use asphalt shovels to spread asphalt or simulated asphalt mix in preparation for compaction • Use asphalt lute to spread asphalt – demonstrate proper selection of toother or straight edge • Use lute or rake to spread asphalt around manholes or other structures • Use wheel barrow to transfer small amounts of asphalt from haul truck to desired location • Coordinate movements with other crew members and ensure ongoing communication • Demonstrate “bumping” the mat and proper techniques for preparing joints • Demonstrate a variety of other common tasks based on site conditions and available equipment, e.g., tiger torch, etc.

Line (GAC): L COMPACTOR OPERATION

Competency: L1 Describe compaction equipment

Objectives

To be competent in this area, the individual must be able to:

- Describe the functions of the compactor and the drum vs. wheel processes of compaction.
- Describe the functions of the compactor and the vibratory vs. static processes of compaction.
- Describe the functions of each phase of a compaction train.

LEARNING TASKS	CONTENT
1. Describe types of Compaction Rollers	• Manufacturers, e.g., CAT, Amman, Bomag, Dynapac • Sized to match use • Production rate determines size • Type of mix (foreman controlled) • Finished product specs • Experience • Drum widths and weight
2. Describe major components of compaction equipment	• Pneumatic (rubber tire) • Never vibration • Adjust tire pressure to affect depth • Combo rubber on one end, steel on other • Vibration option • Steel – vibration (manual) • Diesel powered
3. Describe basic functions of compaction equipment	• Breakdown – mix, time of year • Intermediate • Finish
4. Describe emerging technology	• New in industry – oscillating (side to side) less impact

Line (GAC): L COMPACTOR OPERATION

Competency: L2 Apply compaction work safety procedures

Objectives

To be competent in this area, the individual must be able to:

- Describe how to work safely while working on and around a compactor. This will include understanding key regulations, machine limitations, operating procedures and tasks and the use of PPEs.
- Comply with safety requirements at all times during this course.

LEARNING TASKS	CONTENT
1. Interpret job planning	• Communicate job plan • Path • Pattern • Asphalt thickness • Temperature • Crown • Coordination of asphalt train • Obstacles
2. Describe machine-specific safety hazards	• Dust, pressure (machine malfunctions/kickbacks) blind spots, fumes, hydraulic line, noise • Block outs • ROPS/seatbelts • Working in proximity to paver and other Compaction Rollers – 3 meters is best practice
3. Describe traffic control requirements	• Signage • Signals • Communication
4. Describe PPE	• Workers are responsible to provide general purpose gloves, safety footwear, hard hat - employers are responsible for providing all other safety items • Employers must ensure that PPE is properly worn when required, and is cleaned, inspected, maintained • Employers must ensure that workers are adequately trained: correct use, limitations, maintenance

LEARNING TASKS	CONTENT
5. Describe lockout procedures	• Fuel supply • Battery lockout, etc.
6. Describe how to secure machine for safety	• Vandal proof take ancillary parts off • Park safety block elevated equipment, use parking brakes and shocks

Achievement Criteria

Performance — The learner will demonstrate safe work practices while working with and operating the compaction roller.
Learners will demonstrate safe practices during practical training and on the job.

Conditions — The learner will use:

- Use correct PPE
- Compaction roller

Criteria — The learner will secure the machine for safety including:

- Lock-out procedures
- Chalks and blocks
- Using night switch

The learner will demonstrate safe operating techniques including:

- Demonstrate machine specific safety awareness
- Demonstrate operating procedures
- Demonstrate safe parking practices

Line (GAC): L COMPACTOR OPERATION

Competency: L3 Describe asphalt Compaction Roller operator roles and responsibilities

Objectives

To be competent in this area, the individual must be able to understand the roles and responsibilities of the Compaction Roller crew.

LEARNING TASKS	CONTENT
1. Describe communication techniques	• Coordination and communication among all Compaction Roller operations (two or more)
2. Describe effective rolling pattern	• Compactor and smoothness of asphalt mix
3. Determine pattern to achieve ride planning	• Required ride (cannot adjust after the fact)
4. Describe responsibilities of Compaction Roller operator	• Achieving required density • Finish surface • Asphalt train • Not cutting across the mat too severely • Not stopping with vibrating features • No cut marks, consistency

Line (GAC): L COMPACTOR OPERATION

Competency: L4 Perform Compaction Roller maintenance

Objectives

To be competent in this area, the individual must be able to perform regular daily maintenance on the compactor.

LEARNING TASKS	CONTENT
1. Service lubrication systems (greasing)	• Identify and lube all points • Load grease gun
2. Service air intake system as required (air filters/pre cleaners)	• Restriction indicators • Pre cleaner service • Cleaning vs. replacement, check employer policy (some insist on never using air – replace only) • Use air only with reduced pressure
3. Drain fuel tank sump, water separator fuel filters is equipped	• Drain assess • Water, debris, scale, etc.
4. Add fluids as required	• Identify correct fluids for each component • Add fluids as required • Avoid contamination with dirt, water • Describe 2 sided dipsticks with levels for stopped/idling (CAT engines, transmissions)
5. Inspect and clean components as required	• Radiator and oil cooler • Filter plugs, dipsticks • Battery terminals
6. Perform housekeeping tasks	• Garbage • Excess grease • Maintain sprayers • Cleanliness of tires or drums • Winter – drain all water • Replace/maintain scrapers and coco mat (indicated by use)

Achievement Criteria

Performance	The learner will perform regular daily maintenance on the compaction roller. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will demonstrate all elements of daily maintenance.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist.